Transformative Science and the Big Questions Shaping the Future of Prostate Cancer

A Perspective in Memory of Felix Y. Feng, MD

Kenneth J. Pienta, MD
Donald S. Coffey Professor of Urology, Oncology, and Pharmacology and Molecular Sciences
Director of Research, The James Buchanan Brady Urological Institute
Co-Director, Prostate Cancer Research Program, Sidney Kimmel Comprehensive Cancer Center
The Johns Hopkins University School of Medicine, Baltimore, Maryland, USA

Prepared for the Feng Symposium on Prostate Cancer Revolutions, University of California, San Francisco

Abstract

Felix Y. Feng, MD, died of cancer on December 10, 2024, seven months after he delivered the keynote at the symposium that now carries his name. Across a career that produced more than 500 papers and trained a generation of genitourinary oncologists, he kept returning to five questions that still define the field. Is prostate cancer one disease or many. What actually drives resistance to therapy. How much treatment does each patient really need. Can a machine learn to treat the patient in front of us instead of only the patients behind us. And does the science a person builds outlive the person who built it. This Perspective revisits those five questions through the lens of tumor ecology and evolution, the framework that has organized my own laboratory's work on prostate cancer for three decades. Prostate cancer is not a monolith. It is a population of cell lineages under selection, and the genomic classifiers Feng's laboratory built were early attempts to read that population's trajectory before treatment forces its hand. Resistance is not a switch that fails. It is a Darwinian outcome we help select for every time we treat to progression instead of to control. I argue that the next decade of prostate cancer research should make that ecological view explicit: build classifiers that forecast evolutionary trajectories and not just risk tiers, and build the shared infrastructure that lets a scientist's questions keep being asked after he is gone.

Keywords: prostate cancer, tumor evolution, genomic classifiers, treatment resistance, precision oncology, artificial intelligence, mentorship

A Talk He Never Finished

"When a symposium is named after you, you can pretty much just wax poetic up on stage," Felix Feng told the room on May 18, 2024. "I always tell you to give a crisp talk, one or two major take-home messages. I'm gonna break all those rules, okay?" [1] He was 48 years old, giving the opening keynote at a meeting organized in his honor, seven months before his own cancer would take him. He broke the rules that day because he understood something the rest of us are still catching up to: a career is not a single take-home message. It is a running argument with the disease, conducted one patient, one gene,

one clinical trial at a time, and the argument does not resolve cleanly just because the speaker's time runs out.

Felix Feng spent his career asking five questions, and none of them are settled. I did not know him as an intimate collaborator. I knew him the way most of us in this field did, as the person whose papers you read before writing your own, whose trial designs you borrowed, whose name showed up on the committee that decided whether your idea was fundable. That is its own kind of closeness. Writing this Perspective for the symposium that bears his name, I want to do what he asked of his own trainees and never quite let himself do: give a crisp accounting of what those five questions were, what his laboratory contributed toward answering them, and what is left for the rest of us to finish. I will read his questions through the ecological and evolutionary framework that has organized my own work on this disease, because I think that framework is where his answers were always pointing, whether he used that language or not.

Question One: Is Prostate Cancer One Disease, or Many?

For most of the twentieth century we treated prostate cancer as a single entity distinguished only by grade and stage, a lump to be sized rather than a population to be characterized. Feng's laboratory spent a decade taking that lump apart. Working from Arul Chinnaiyan's group at the University of Michigan, his team ran RNA sequencing across more than a hundred prostate tissues and found dozens of long non coding RNAs that conventional exon arrays had never seen, among them PCAT-1, a prostate-restricted transcript that turned out to regulate BRCA2 and to drive MYC expression through its 3′ UTR [2,3]. A companion effort identified SChLAP1, overexpressed in roughly a quarter of prostate cancers, which promotes invasion by directly antagonizing the SWI/SNF chromatin remodeling complex [4]. In a validation cohort of more than a thousand men, high SChLAP1 expression independently predicted ten year metastasis with an odds ratio above two [5]. These were not incremental biomarkers. They were evidence that "prostate cancer" was always a label of convenience covering multiple molecular diseases that happened to share an organ of origin.

The clinical payoff came when his group moved from cataloguing subtypes to acting on them. Working with genomic classifier data derived from more than a thousand patients, his team showed that luminal B tumors carry the worst prognosis and are also the only subtype that benefits from postoperative androgen deprivation, a fact invisible to anyone looking at grade and stage alone [6]. Not every high-risk tumor is high risk for the same reason, and not every patient who looks similar on a pathology report will respond to the same drug. **The unit of prostate cancer research is not the organ. It is the lineage.** Feng's own framing of this, delivered in an interview two years before he died, was direct: not all prostate cancers are the same, and the field needed to stop putting them into one disease [7]. His institution's Benioff Initiative, which he directed, states the same idea as a founding premise: the goal is to move from a one size fits all approach to a strategy where therapy is informed by the molecular mechanisms governing each patient's individual cancer [8].

I would push this one step further than Feng's own papers did, because it is where my own field, cancer ecology, has spent its energy. A luminal B tumor is not a fixed category any more than a wolf population and a coyote population are fixed categories. They are population states, snapshots of a lineage under selection, and the same tumor can drift from one state toward another as treatment changes the fitness landscape it lives in. The genomic classifiers Feng's laboratory built are extraordinary because they gave us a first, coarse map of that landscape. The next generation of classifiers needs to do something his did not yet attempt: track how a given tumor's position on that map moves over the course of treatment, not just where it starts.

Question Two: What Actually Drives Resistance?

Every oncologist has watched a drug that worked stop working. Feng's laboratory spent as much energy explaining why as it did classifying who. The mechanistic thread runs from a 2011 paper showing that the TMPRSS2:ERG fusion protein physically recruits PARP1 and DNA-PKcs, providing the rationale for testing PARP inhibitors in ETS fusion positive prostate cancer years before PARP inhibitors reached the clinic in this disease [9], through a 2013 paper describing a feedback circuit in which androgen receptor signaling and DNA-PK mediated repair sustain each other [10], to a 2015 paper establishing that DNA-PKcs acts as a transcriptional driver of metastatic progression in its own right, independent of its repair function [11]. Taken together, this body of work reframed DNA repair machinery in prostate cancer from a downstream bystander into an upstream engine of both androgen signaling and metastatic spread.

Resistance, in this framing, is not a mechanical failure. **It is a selection event, and we are the ones doing the selecting.** Every course of androgen deprivation, every line of a PARP inhibitor, every round of radiation applies a pressure that some fraction of the tumor's cell population survives, and survival under pressure is precisely what evolution rewards. Feng's group gave this idea a direct clinical demonstration when they used circulating cell free DNA to detect BRCA2 reversion mutations in men progressing on PARP inhibitors and found seven distinct reversion alleles in patients where a single tissue biopsy had only turned up two [12]. A tumor does not develop one resistance mechanism. It runs several experiments in parallel and lets the best one win, and a single needle biopsy will always undersample the experiment.

This is where I think the field, including my own laboratory, still owes Feng's questions a fuller answer. We have gotten very good at detecting resistance after it emerges. We remain much worse at predicting which resistance route a given tumor is most likely to take before we commit to a therapy that will select for it. Feng himself named the destination plainly, in a conversation about the long arc of the field: understanding what drives resistance and how to overcome it, he said, may be the key to curing cancer, and thirty years from now we should aim either to cure all cancers or to make it possible for patients to live with cancer without dying of it [13]. He called that the holy grail and said bluntly that there is no other reason to be in science. I do not think that claim was oversized. I think it was exactly sized, stated by someone who had spent a career narrowing the gap between the two.

Question Three: How Much Treatment Does This Patient Actually Need?

Overtreatment and undertreatment are two names for the same failure: giving a population level answer to an individual level question. Feng put this in his own words in one of his last public interviews. Across the landscape of prostate cancer care, he said, some patients are overtreated, some are undertreated, and some get exactly the right treatment, and his goal was to shrink the first two groups and grow the third [14]. That is not a hedge. It is a research program stated as a sentence.

His clearest attempt to execute that program was the genomic classifier work built on the NRG Oncology RTOG 9601 trial, where he was first author on the definitive validation study. Among men randomized to salvage radiation with or without bicalutamide after prostatectomy, the twelve year absolute overall survival benefit from adding hormone therapy was 2.4 percent in men with a low genomic classifier score and 8.9 percent in men with a high score. Restricted to men who received early salvage radiation at a low PSA, the effect reversed: negative 7.8 percent in low score patients, positive 4.6 percent in high score patients, meaning the men most likely to be reflexively given extra hormone therapy were, on average, the ones being harmed by it [15]. A related classifier built on the same underlying platform, PORTOS, went further and asked not whether a patient would fail without radiation but whether radiation itself would work for this specific tumor. Among high PORTOS score patients, radiation was associated with a ten year metastasis rate of five percent versus sixty three percent without it. Among low PORTOS score patients, radiation conferred no measurable benefit at all [16]. That second finding deserves more attention than it usually gets. A test that only ever tells you a patient is at higher risk is a prognostic test. A test that tells you a specific treatment will not help a specific patient is a permission slip to spare that patient a treatment altogether, and permission slips to do less are far rarer in oncology than reasons to do more.

The clinical trials his committee at NRG Oncology built after these findings, trials with names like PREDICT-RT and GUIDANCE, are structured explicitly around this question. Intensify treatment for the genomic subgroup that standard risk categories under-treat. De-intensify for the subgroup they over-treat. Let the classifier, not the stage, make that call [17]. This is the least glamorous kind of consequential science, because its success is measured in treatments withheld rather than treatments delivered, and a withheld treatment produces no headline. It should produce one anyway.

Question Four: Can a Machine Learn to Treat the Patient in Front of Us?

Feng's laboratory did not stop at hand curated genomic signatures. In one of his last major papers, built with the computer scientist Andre Esteva, his group trained a multimodal deep learning model on digitized pathology and clinical data from five phase III randomized trials spanning ten to twenty years of follow up, and the resulting classifier predicted five and ten year metastasis better than PSA and Gleason score, the two variables that have anchored prostate cancer risk stratification for forty years [18]. Feng was explicit about why he trusted this particular data source over the much larger but unrandomized

real world datasets that dominate most AI oncology efforts: phase III trials are randomized, which makes their outcomes believable in a way that retrospective cohorts are not, and pathology slides prepared decades ago are prepared essentially the same way slides are prepared today, which means archival trial material can be digitized and mined without becoming obsolete [19]. That is a methodological discipline that a lot of AI enthusiasm in oncology still lacks. The most valuable training data is not the biggest. It is the data whose provenance you can defend in front of a skeptical colleague.

He was equally direct about where these tools fail, and here his answer is the part of his legacy I would most like this symposium to inherit intact. An AI tool, he said, works primarily in the populations it was built from, and despite years of genomic testing in prostate cancer, the number of African American men who had undergone that testing remained strikingly low relative to their share of the disease burden. His response was not a disclaimer buried in a methods section. He built his classifiers on cohorts that were roughly twenty percent African American, validated them in matched populations, and reported that the resulting tool performed equally well across groups, concluding that the more inclusive the training data, the better the classifier performs for everyone [20]. **Health equity, in his framing, was not an ethical addendum to good machine learning. It was a technical precondition for it.** A model trained on one population and deployed on another is not a shortcut. It is a different, unvalidated instrument wearing the same name, and pretending otherwise is a category of error we should stop tolerating.

I want to be honest about the size of this claim, because the field's appetite for AI has outrun its patience for validation. A model trained on five phase III trials is a genuine advance over PSA and Gleason score. It is not yet a replacement for the randomized trial that will need to test whether acting on its output changes outcomes. Feng understood that distinction, which is why his classifier work kept returning to trial platforms rather than retrospective registries. The next big question for this line of work is not whether a model can predict outcome. Several already can. It is whether a health system can be organized to act differently, and defensibly, on that prediction, in populations that look like the country rather than like the training set.

Question Five: Does the Science Outlive the Scientist?

Every obituary I read about Felix Feng mentioned his publication count. Almost none of them led with it. They led instead with what Eric Small called his ability to build communities of people, collaborations, and teams like no one else, and with what Carissa Chu, one of his mentees, described as a habit of climbing quickly toward the top and then reaching back for the youngest person in the room [21]. Feng said it himself, more plainly than most of us manage: the magnitude of what a scientist accomplishes is not measured only in what that scientist has done, but in the collective effort of everyone helped along the way, and his greatest contribution to the field would be the careers he helped build rather than the papers he authored [22].

That is not modesty. It is an accurate description of what happened after he died. Within months, the American Cancer Society and ASTRO endowed a clinician scientist development grant in his name. ASCO's Conquer Cancer foundation endowed a young investigator award in prostate cancer, awarded for the first time in 2025 [23]. The NRG Oncology genitourinary committee he chaired for six years continued running the biomarker guided trials he designed, including one, BALANCE, that reported the first prospectively validated predictive biomarker in prostate cancer at ASTRO's 2025 meeting, nearly a year after his death [24]. The prostate cancer program he co-led secured a new Specialized Program of Research Excellence grant from the National Cancer Institute the same month it lost its co-director [25]. And this symposium exists at all because, as one account of the week he died put it, more than two hundred people gathered on short notice to make sure his name would be attached to a meeting that kept happening [26].

I read all of that not as a story about one remarkable man but as a natural experiment in institutional design. Feng did not just discover things. He built the infrastructure, the training grants, the multi-institutional trial networks, the named professorships, the annual meetings, that let his questions keep being asked by other people using his tools after he was no longer the one asking them. That is the least studied and most important skill in academic medicine, and it is the one talent no impact factor measures. A tumor's evolutionary trajectory outlasts any single course of therapy. A scientist's intellectual trajectory, it turns out, can be made to outlast the scientist, but only if he spends deliberate effort building the scaffolding for that to happen while he is still here to build it. Felix Feng spent effort on that scaffolding as if he already knew he would need it.

What Is Left to Finish

I opened with Feng breaking his own rule about the crisp take home message, so I will honor the rest of his instruction and try to give one. **Prostate cancer research over the next decade should stop treating heterogeneity, resistance, overtreatment, algorithmic bias, and scientific succession as five separate problems, because Feng's own body of work shows they are one problem viewed from five angles.** A tumor is a population under selection. Every classifier we build should be judged by whether it predicts that population's next move, not just its current label. Every trial we design should be judged by whether it tells us who can safely receive less, not only who needs more. Every algorithm we deploy should be judged against populations it was never trained on before we let it near a patient who looks like one. And every laboratory we run should be judged, in part, by what continues to run in it after we are gone.

None of the five questions Feng asked are closed. That is not a failure of his twenty year career. It is the honest shape of a field that is still young enough to have its foundational questions asked by a single working scientist rather than settled by consensus. The keynote he gave in May of 2024 ended, by every account, without a crisp final slide. Seven months later he was gone, and the questions were still open, sitting exactly where he had left them for the rest of us to pick up. That is what this symposium is for.

That is what this Perspective is for. The work is not to memorialize a body of answers. It is to keep asking his questions with the same discipline he brought to them, using tools he would not have had time to build himself.

Acknowledgments and Disclosures

The author declares no financial conflicts of interest related to this work. This Perspective was prepared to accompany the author's keynote address at the Feng Symposium on Prostate Cancer Revolutions and draws on the publicly available scientific record, institutional obituaries, and interviews of Felix Y. Feng, MD, cited throughout. The author is grateful to the UCSF Benioff Initiative for Prostate Cancer Research and the organizers of the Feng Symposium for the opportunity to reflect on Dr. Feng's scientific legacy in the company of his colleagues, trainees, and friends.

References

1. The Cancer Letter. Felix Feng: "When a symposium is named after you, you can pretty much just wax poetic up on stage." Jan 24, 2025. https://cancerletter.com/in-the-archives/20250124_8/

2. Prensner JR, Iyer MK, Balbin OA, et al. Transcriptome sequencing across a prostate cancer cohort identifies PCAT-1, an unannotated lincRNA implicated in disease progression. Nat Biotechnol. 2011;29(8):742-749. https://www.nature.com/articles/nbt.1914

3. Prensner JR, Chen W, Han S, et al. PCAT-1, a long noncoding RNA, regulates BRCA2 and controls homologous recombination in cancer. Cancer Res. 2014;74(6):1651-1660. https://pubmed.ncbi.nlm.nih.gov/24473064/

4. Prensner JR, Iyer MK, Sahu A, et al. The long noncoding RNA SChLAP1 promotes aggressive prostate cancer and antagonizes the SWI/SNF complex. Nat Genet. 2013;45(11):1392-1398. https://www.nature.com/articles/ng.2771

5. Mehra R, Udager AM, Ahearn TU, et al. RNA biomarkers associated with metastatic progression in prostate cancer: a multi-institutional high-throughput analysis of SChLAP1. Lancet Oncol. 2014;15(13):1469-1480. https://pubmed.ncbi.nlm.nih.gov/25456366/

6. Zhao SG, Chang SL, Erho N, et al. Associations of Luminal and Basal Subtyping of Prostate Cancer With Prognosis and Response to Androgen Deprivation Therapy. JAMA Oncol. 2017;3(12):1663-1672. https://pubmed.ncbi.nlm.nih.gov/28494073/

7. Grand Rounds in Urology / Prostatepedia. Frontiers in Prostate Cancer Genomics. Interview with Felix Feng, MD. https://grandroundsinurology.com/frontiers-in-prostate-cancer-genomics/

8. UCSF Benioff Initiative for Prostate Cancer Research. Research overview. https://cancer.ucsf.edu/research/benioff-prostate-initiative

9. Brenner JC, Ateeq B, Li Y, et al. Mechanistic rationale for inhibition of poly(ADP-ribose) polymerase in ETS gene fusion-positive prostate cancer. Cancer Cell. 2011;19(5):664-678. https://pubmed.ncbi.nlm.nih.gov/21575865/

10. Goodwin JF, Schiewer MJ, Dean JL, et al. A hormone-DNA repair circuit governs the response to genotoxic insult. Cancer Discov. 2013;3(11):1254-1271. https://pubmed.ncbi.nlm.nih.gov/24027197/

11. Goodwin JF, Kothari V, Drake JM, et al. DNA-PKcs-Mediated Transcriptional Regulation Drives Prostate Cancer Progression and Metastasis. Cancer Cell. 2015;28(1):97-113. https://pubmed.ncbi.nlm.nih.gov/26175416/

12. Quigley DA, Dang HX, Zhao SG, et al. Analysis of circulating cell-free DNA identifies multi-clonal heterogeneity of BRCA2 reversion mutations associated with resistance to PARP inhibitors. Cancer Discov. 2017;7(9):999-1005. https://pmc.ncbi.nlm.nih.gov/articles/PMC5581695/

13. The V Foundation. How Dr. Felix Feng Is Aiming for the Holy Grail in Prostate Cancer Research. https://www.v.org/story/how-dr-felix-feng-is-aiming-for-the-holy-grail-in-prostate-cancer-research/

14. UroToday. Personalizing Prostate Cancer Care with AI: Felix Feng. Nov 2, 2022. http://www.urotoday.review/video-lectures/prostate-cancer/video/2914-personalizing-prostate-cancer-care-with-ai-felix-feng.html

15. Feng FY, Huang HC, Spratt DE, et al. Validation of a 22-Gene Genomic Classifier in Patients With Recurrent Prostate Cancer: An Ancillary Study of the NRG/RTOG 9601 Randomized Clinical Trial. JAMA Oncol. 2021;7(4):544-552. https://jamanetwork.com/journals/jamaoncology/fullarticle/2776225

16. Zhao SG, Chang SL, Spratt DE, et al. Development and validation of a 24-gene predictor of response to postoperative radiotherapy in prostate cancer: a matched, retrospective analysis. Lancet Oncol. 2016;17(11):1612-1620. https://pubmed.ncbi.nlm.nih.gov/27743920/

17. NRG Oncology. PREDICT-RT (NRG-GU009) and GUIDANCE (NRG-GU010) protocol pages. https://www.nrgoncology.org/PREDICT-RT

18. Esteva A, Feng J, van der Wal D, et al. Prostate cancer therapy personalization via multi-modal deep learning on randomized phase III clinical trials. NPJ Digit Med. 2022;5(1):71. https://pubmed.ncbi.nlm.nih.gov/35676445/

19. UroToday. Personalizing Prostate Cancer Care with AI: Felix Feng. Nov 2, 2022 (as in reference 14). http://www.urotoday.review/video-lectures/prostate-cancer/video/2914-personalizing-prostate-cancer-care-with-ai-felix-feng.html

20. UroToday. Personalizing Prostate Cancer Care with AI: Felix Feng. Nov 2, 2022 (as in reference 14). http://www.urotoday.review/video-lectures/prostate-cancer/video/2914-personalizing-prostate-cancer-care-with-ai-felix-feng.html

21. The Cancer Letter. In Memoriam: Felix Y. Feng, MD. Jan 24, 2025. https://cancerletter.com/obituary/20250124_4/

22. The ASCO Post. Felix Feng Strives for Balance Between Work and Family Life. Jan 25, 2020. https://ascopost.com/issues/january-25-2020/felix-feng-strives-for-balance-between-work-and-family-life/

23. American Society for Radiation Oncology. The ACS-ASTRO Felix Feng, MD Clinician Scientist Development Grant Award. https://www.astro.org/about-astro/board-and-leadership/acs-astro-felix-feng-grant-award

24. Urology Times. PAM50 biomarker is predictive of hormone therapy benefit in post-prostatectomy setting (NRG-GU006/BALANCE, ASTRO 2025). https://www.urologytimes.com/view/pam50-biomarker-is-predictive-of-hormone-therapy-benefit-in-post-prostatectomy-setting

25. UCSF Helen Diller Family Comprehensive Cancer Center. UCSF Prostate Cancer Program Awarded SPORE Grant by National Cancer Institute. Sept 3, 2024. https://cancer.ucsf.edu/news/2024/09/03/ucsf-prostate-cancer-program-awarded-spore-grant-by-national-cancer-institute

26. The Cancer Letter. In Memoriam: Felix Y. Feng, MD (continued). Jan 24, 2025. https://cancerletter.com/obituary/20250124_5/